

Are You Getting What You Pay For?

Auditing Model Substitution in LLM APIs

Why Software Tests Fail and Hardware Attestation Wins

arXiv: 2504.04715

The trust gap in commercial LLM APIs: providers may covertly swap models, and software-based detection cannot reliably catch them.

The Model Substitution Problem: Economic Incentives Drive Covert Swaps

Commercial LLM API providers charge for specific models, but the black-box nature of APIs means users have no guarantee the advertised model is actually being served. Providers face strong economic incentives to covertly substitute cheaper alternatives to reduce GPU costs while maintaining pricing.

Formal Hypothesis Testing

H_0 : Samples come from $P_{\text{spec}}(y|x)$ — the specified model

H_1 : Samples come from $P_{\text{actual}}(y|x)$ — a substitute model

Can an auditor reliably reject H_0 ?

⚠ Economic Incentive

GPU compute is the dominant cost in serving LLMs. Replacing a 70B model with an 8B model or a quantized variant can reduce inference costs by 5-10x while maintaining pricing, creating a direct profit motive for covert substitution.

Users pay for 70B-class performance but may receive 8B-class output — and cannot verify the difference from black-box queries alone.

1. Smaller Model

e.g., 8B instead of 70B

2. Quantized Variant

FP8/INT8 instead of FP16

3. Fine-tuned Version

Updated or retrained model

4. Different Family

Entirely different architecture

FP8 Quantization Preserves Most Accuracy — Making Substitution Hard to Detect

Performance drops from FP8 quantization are small, often within error bars — especially on MMLU and GSM8K. An auditor cannot reliably distinguish FP8 from FP16.

Model	MMLU	GSM8K	MATH	GPQA Diamond
Llama-3-8B-Instruct	62.69 ± 0.18	61.14 ± 3.47	20.65 ± 3.43	22.62 ± 0.22
Llama-3-8B-Instruct-FP8	62.43 ± 0.26	60.90 ± 4.10	14.91 ± 2.49	20.14 ± 0.24
Llama-3-70B-Instruct	78.05 ± 0.08	88.06 ± 1.44	35.69 ± 1.33	29.60 ± 0.51
Llama-3-70B-Instruct-FP8	77.88 ± 0.13	87.35 ± 1.24	35.75 ± 1.16	33.12 ± 0.30
Gemma-2-9b-it	71.86 ± 0.08	81.80 ± 1.35	33.41 ± 0.28	28.64 ± 2.97
Gemma-2-9b-it-FP8	71.92 ± 0.11	79.41 ± 1.14	32.53 ± 0.34	27.81 ± 3.22
Qwen2-72B-Instruct	82.18 ± 0.08	86.72 ± 1.00	37.39 ± 1.41	29.93 ± 2.71
Qwen2-72B-Instruct-FP8	81.98 ± 0.08	86.82 ± 0.97	37.67 ± 1.39	31.08 ± 1.96
Mistral-7B-Instruct-v0.3	59.15 ± 0.10	35.90 ± 4.54	8.94 ± 1.24	21.60 ± 0.17
Mistral-7B-Instruct-v0.3-FP8	58.77 ± 0.13	32.20 ± 4.02	7.68 ± 1.23	22.72 ± 0.19

Key Insight: For most models, FP8 scores fall within the error bars of the original — an auditor comparing benchmark scores alone cannot distinguish quantization from natural variance.

Higher Temperature Amplifies Performance Gaps but Increases Noise

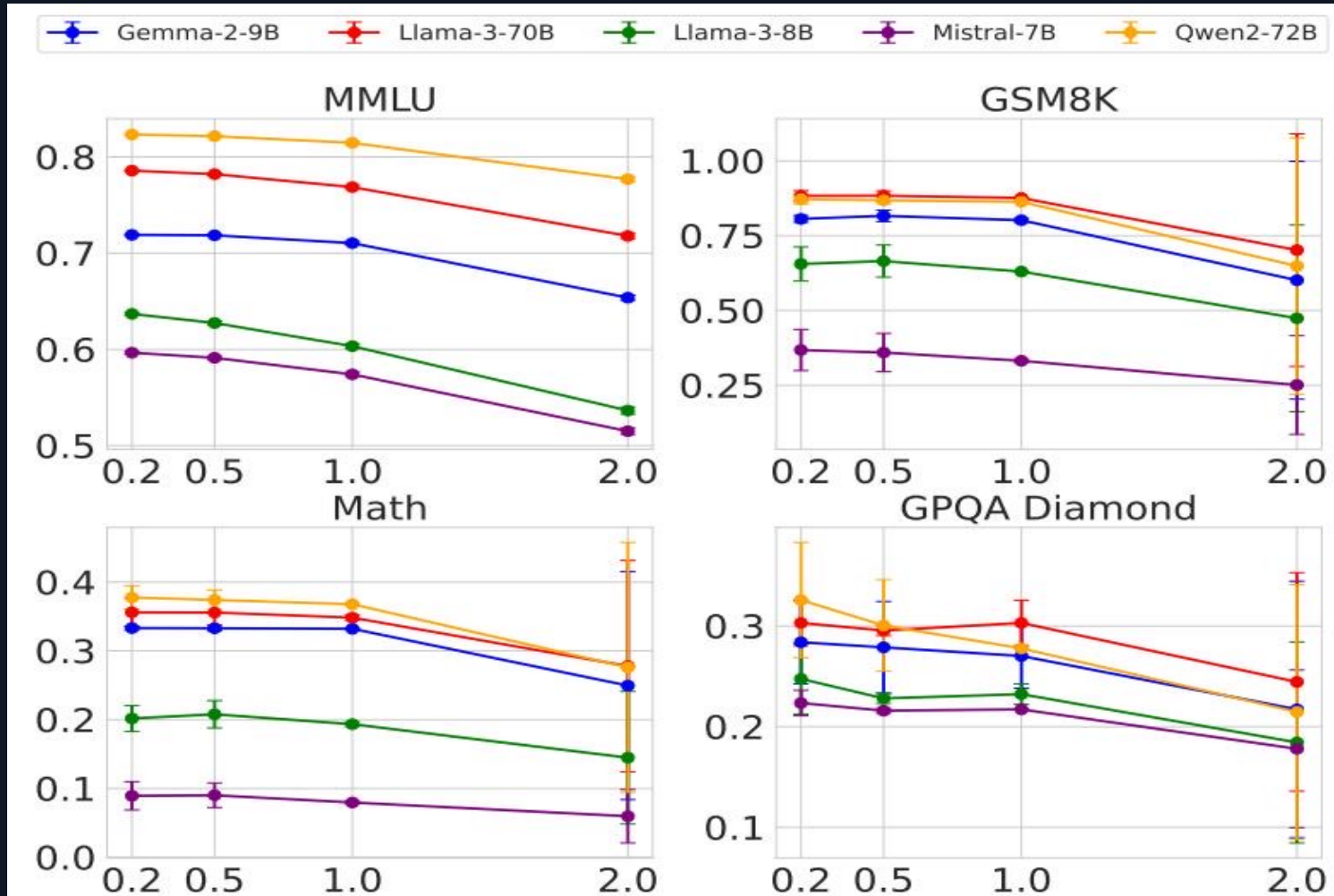


Figure: Benchmark accuracy vs. temperature for five models across four tasks (MMLU, GSM8K, MATH, GPQA Diamond)

Observations

- Temperature 0.2–0.5: Models cluster tightly; original vs. FP8 differences are minimal.
- Temperature 1.0: Performance gaps begin to widen but variance increases substantially.
- Temperature 2.0: All models degrade significantly. The gap between original and substitute narrows or becomes noisy.
- Implication: Higher temperature — a common audit strategy — introduces enough noise to defeat statistical detection rather than help it.

Log Probability Fingerprinting Fails Due to Inference Nondeterminism

Model	BERT Acc	T5 Acc	GPT2 Acc	LLM2Vec Acc
Llama3-70B-Instruct-FP8	50.55	50.10	50.45	49.90
Llama3-70B-Instruct-INT8	51.60	49.90	51.30	50.25
Gemma2-9b-it-FP8	49.95	50.30	51.20	49.80
Gemma2-9b-it-INT8	49.00	49.70	51.65	49.55
Mistral-7b-v3-Instruct-FP8	50.55	49.75	48.70	48.75
Mistral-7b-v3-Instruct-INT8	49.70	50.75	50.50	51.15
Qwen2-72B-Instruct-FP8	50.05	50.25	48.75	49.80
Qwen2-72B-Instruct-INT8	50.75	50.55	49.45	50.20

Why Fingerprinting Breaks

- Same model (Llama-3-8B) produces different log probabilities across inference frameworks.
- Software variation: transformers 4.40 vs 4.50, vLLM 0.6.2 vs 0.8.2 yield different outputs.
- Hardware variation: A100 vs H100 GPUs introduce numerical differences.
- An auditor cannot distinguish between a legitimate model on different infrastructure and an actual substitution.

Figure: Log probability variation across inference frameworks and hardware for Llama-3-8B

Randomized Routing at Low Rates Makes Statistical Detection Query-Prohibitive

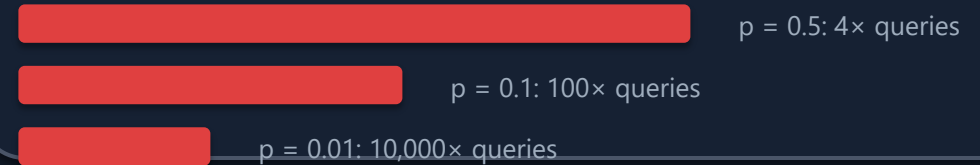
The provider routes only a fraction p of queries to the cheaper substitute model, serving the genuine model for the remaining $(1-p)$ queries. As p approaches 0, the mixed distribution converges to the specified model's distribution, requiring exponentially more queries for detection.

Mixed Distribution

$$P_{\text{mixed}}(y|x) = p \cdot P_{\text{sub}}(y|x) + (1-p) \cdot P_{\text{spec}}(y|x)$$

As $p \rightarrow 0$: $P_{\text{mixed}} \rightarrow P_{\text{spec}} \Rightarrow$ Detection requires $O(1/p^2)$ queries

Query Cost Explosion



Adversarial Strategy: Benchmark Evasion

Providers can detect known benchmark prompts (MMLU, GSM8K, etc.) and route those queries to the genuine model, while serving substitutes for all other traffic.

Auditors relying on standard benchmarks will see normal performance, even though real users receive substitute-quality output.

This makes any benchmark-based audit trivially gameable.

Combined: randomized routing + benchmark evasion = software-only detection is fundamentally query-prohibitive

Software-Only Auditing Methods Are Fundamentally Unreliable

Method Type	Weakness	Practical Limitation
Text-Based Statistical Tests (e.g., MMD)	Require impractically large query budgets; fail against subtle substitutions where $P_{\text{spec}} \approx P_{\text{sub}}$	FP8 quantization drops are within error bars — statistical tests cannot distinguish them from natural variance
Log-Probability Methods	Defeated by production-level inference nondeterminism across software versions and hardware	Same model yields different log probs on A100 vs H100, transformers vs vLLM — auditor cannot distinguish framework noise from substitution
Adversarial Countermeasures	Providers actively evade audits via randomized routing and benchmark detection	Randomized substitution ($p \rightarrow 0$) requires $O(1/p^2)$ queries; benchmark evasion makes standard audits gameable

Verdict: No software-only method provides reliable detection

Each method fails independently: statistical tests lack power against near-identical distributions, log-probability fingerprinting collapses under real-world nondeterminism, and adversarial providers can actively evade any detection strategy. These are not implementation flaws — they are fundamental information-theoretic limitations of black-box auditing.

Trusted Execution Environments Provide Provable Cryptographic Model Integrity

How TEEs Guarantee Model Integrity

- ✓ Hardware-secured enclaves provide cryptographic attestation of the code and data being executed.
- ✓ The provider can prove that specific model weights are loaded and the correct inference code is running.
- ✓ Attestation is verified by the client — not the provider — eliminating the trust gap.
- ✓ Only modest performance overhead compared to untrusted inference.

TEE vs. Alternatives

Property	Software	ZKP	TEE
Provable	✗	✓	✓
Efficient	✓	✗	✓
Robust to Evasion	✗	✓	✓

ZKPs are computationally prohibitive for billion-parameter models. TEEs are practical and deployable today.

Practical Deployment: NVIDIA Confidential Computing

- NVIDIA H100 GPUs support confidential computing mode, enabling TEE-protected inference on current hardware.
- Providers generate cryptographic attestation reports that clients can verify independently.
- No changes to model architecture required — any model that runs on the GPU can be attested.
- The attestation covers the entire inference pipeline: model weights, inference code, and runtime configuration.

Key Takeaways: Close the Verification Gap with Hardware Attestation

1

Model substitution is economically motivated and practically undetectable via software

Providers save 5–10x on GPU costs by swapping models. The black-box nature of APIs gives users no way to verify what they receive.

2

FP8 quantization barely affects benchmarks

Drops are within error bars on MMLU and GSM8K. An auditor comparing scores alone cannot detect the swap.

3

Inference nondeterminism kills log-probability fingerprinting

Same model produces different log probs across frameworks (transformers vs vLLM) and hardware (A100 vs H100).

4

Randomized substitution defeats statistical tests

As $p \rightarrow 0$, the mixed distribution converges to the specified model's, requiring exponentially more queries.

5

TEEs are the only currently viable path to verifiable LLM APIs

Hardware attestation provides provable, efficient, and robust guarantees. NVIDIA confidential computing is deployable today.

Call to Action: The LLM community must adopt hardware attestation as the standard for API integrity verification. Software-only audits provide a false sense of security.